Srinivasa Manikanta Rajapantula

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SUMMARY

A passionate Electrical and Electronics Engineering student with hands-on experience in embedded systems, IoT, and full-stack development, driven by a strong interest in applying engineering principles to solve real-world problems.

SKILLS

Embedded Systems & Hardware: ESP32, ESP8266, Arduino, Communication Protocols (UART, I2C, SPI, MQTT), Sensors & Actuators

Internet of Things: IoT System Design, Cloud Integration, Real-time Monitoring

Programming Languages: C, Embedded C, Assembly, Python, MATLAB

Power Systems: Power Generation, Transmission & Distribution, Smart Grid Concepts, MATLAB Simulation

Power Electronics: Converters, Inverters, Motor Drives, Thyristors, Choppers, PWM Control

Electrical Machines: DC & AC Machines, Transformers, Electrical Motors

Additional Skills *(AI-assisted)*: Web Development (React.js, Next.js), Database (Firebase, Firestore), App Development (React Native, Expo)

PROJECTS

Project	Tech Stack	Highlight	Link
Single Axis Solar Tracker	LDR, Relay, Gear Motor	Pure relay logic, ~30% efficiency gain	—
Automatic Exhaust Fan	MQ-2, Arduino, Relay	Auto-off timer, manual override	autoexhaustfan.rsmk.co.in
Automatic Streetlight System	IC 555, LDR, Solar Panel	Fully solar powered, mini model	—
Solar-Powered Dewatering System	Arduino, LDR, Soil Moisture, Solar	SIH project, college level, prototype built	spds.rsmk.me
Firefighter Safety Device	ESP32, Firebase, React/Vite	A-HACKS 2nd Prize, Hardware Category	sfmd.rsmk.co.in
ColorOhm	React Native, Expo	Published Android app, GitHub v1.0.0	colorohm.rsmk.me

EXPERIENCE

Industrial Trainee — Electrical & Electronics

Coromandel International Limited, Visakhapatnam | *6th Semester, Diploma — 2024*

- Observed 132kV/11kV substation operations including power distribution across multiple plant sections
- Studied Turbo Generator (TG) station — 5MW power generation from waste heat of sulphuric acid plant
- Learned operation of Diesel Generator sets (DG1–4MW, DG2–4MW, DG3–6MW) and emergency DG systems
- Understood Motor Control Center (MCC) with DOL starters, contactors, and overload relays
- Studied substation components — Circuit Breakers, CTs, PTs, Lightning Arresters, Isolators
- Gained exposure to electrical safety practices and Maintenance Work

EDUCATION

B.Tech — Electrical & Electronics Engineering | CGPA: 7.5
Andhra Loyola Institute of Engineering & Technology, Vijayawada / *2024 – 2027 (Pursuing)*

Diploma — Electrical & Electronics Engineering | 75%
Government Polytechnic Srikakulam / *2021 – 2024*

SSC (Class 10) | 542/600 (90.3%)
ZPHS Murapaka, AP State Board / *2021*

CERTIFICATIONS

- Drone Technology
- Building IoT Solutions with Embedded Systems
- M-botix Training
- Introduction to Internet of Things (IoT) — NPTEL
- 3D Printing
- Electric Vehicle Technology
- Industrial Internet of Things (IIoT)

ACHIEVEMENTS

- 🥇 Department 1st — Firefighter Safety Device Project
- 🥈 2nd Place — A-HACKS College Level Hackathon, Hardware Category
- 🥇 1st Place — IEI Student Chapter Paper Presentation